

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Information Technology) SEMESTER - 6 Summer 2025 (Regular)

Course :Bachelor of Technology (Information Technology) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BITTOE604A - COMPILER DESIGN

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 Pick up the wrong statement from the following
 - a. A token is pair consisting of a token name and an attribute value.
 - b. The token name is an abstract symbol representing a kind of lexical unit.
 - c. Lexeme is a sequence of characters in the source program that matches the pattern for a token.
 - d. Scanning consists of the simple processes that require tokenization of input.
- 2 Consider the string banana then pick the wrong statement/s from following?
 - a. ban, banana and ϵ are the prefixes of given string.
 - b. nana, banana and ϵ are suffixes of given string.
 - c. baan is substring of the given string
 - d. None of the above.
- 3 The top-down parser can't do the following
 - a. Find the leftmost derivation for an input string.
 - b. Recursive descent parsing is used to parse the input string without backtracking.
 - c. Use of lookahead 1 at each step to make the parsing decision
 - d. None of the above.
- 4 Handle pruning is
 - a. It's a process where, if a grammar is unambiguous then every right sentential form of the grammar has exactly one handle.
 - b. It's method that uses a handle as a substring that matches with the head of the production for reducing the substring.
 - c. It uses the reduction whose reduction represents the one step along the reverse of rightmost derivation.
 - d. Both a and b.
- 5 Syntax Directed Definition isn't
 - a. A Context-Sensitive Grammar with attribute and values.
 - b. It specifies the values of attributes associating with semantic rules with the grammar productions.
 - c. It having Inherited and Synthesized attributes.
 - d. An SDD involves only synthesized attributes known as S-attributed.

- 6 Pick-up the not correct statement from the following.
- Dependency graph shows the flow of information among the attribute instances in a particular parse tree.
 - Dependency graph shows an edge from one attribute instance to another means that the value of the second is needed to compute after the first.
 - In L-attributed definition, each attribute must be either synthesized or inherited.
 - None of the above.
- 7 Intermediate Code Generator supports for
- Static checking includes type checking which ensures that operators are applied to compatible operands.
 - High level intermediate representation and Low level intermediate representation, where high level IR are close to target machine and Low level IR close to source language.
 - DAG for finding common sub-expression.
 - None of the above.
- 8 Which isn't correct about three address code
- Contains at most one operator at RHS of an instruction.
 - Uses the temporary variables that are compiler generated temporary names.
 - It is built from only concept of Addresses.
 - Addresses may be name, constant.
- 9 Basic blocks are
- Partition of sequence of three address instruction.
 - Sequence of instruction that are always executed separately.
 - Contains the leader which will be the any instruction that is the target of any conditional or unconditional jump.
 - Consist of Instructions which having the leaders.
- 10 Which option isn't correct for the following grammar?
- $$S \rightarrow ACB|CB B \quad A \rightarrow da|BC \quad B \rightarrow g|\epsilon \quad C \rightarrow h|\epsilon$$
- Where S is start symbol and ϵ is an empty string
- $FIRST(A) = \{d, g, h\}$ & $FOLLOW(B) = \{a, h, g, \$\}$
 - $FIRST(B) = \{g, \epsilon\}$ & $FOLLOW(A) = \{h, g\}$
 - $FOLLOW(C) = \{g, h, b, \$\}$ & $FIRST(S) = \{d, h, b, g\}$
 - All the above.
- 11 Consider the following grammar and pick-up the correct option
- $$S \rightarrow A \quad A \rightarrow Bb|Cd \quad B \rightarrow aB|\epsilon \quad C \rightarrow cC|\epsilon$$
- Where S is start symbol and ϵ is an empty string
- $FIRST(S) = \{a, b, c, d\}$ and Grammar is LL(1) conflict.
 - $FIRST(B) = \{a, \epsilon\}$ and Grammar is LL(1) conflict.
 - $FIRST(A) = \{a, d, c, b\}$ and Grammar isn't LL(1) conflict.
 - $FIRST(S) = \{a, b, c, d, \epsilon\}$ and Grammar isn't LL(1) conflict.
- 12 Which statement/s is/are correct for Peephole Optimization?
- Peephole optimization is done by examining the sliding window of target instruction.
 - If possible replace the instruction sequences within the peephole by shorter or faster sequence.
 - Peephole optimization can use the Flow-of-control optimizations characteristics.
 - All the above.

Q2. Solve the following.

A) Explain the role of lexical analysis with block diagram.

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B) Explain Lexical-Analyzer generator LEX with its structure definition.

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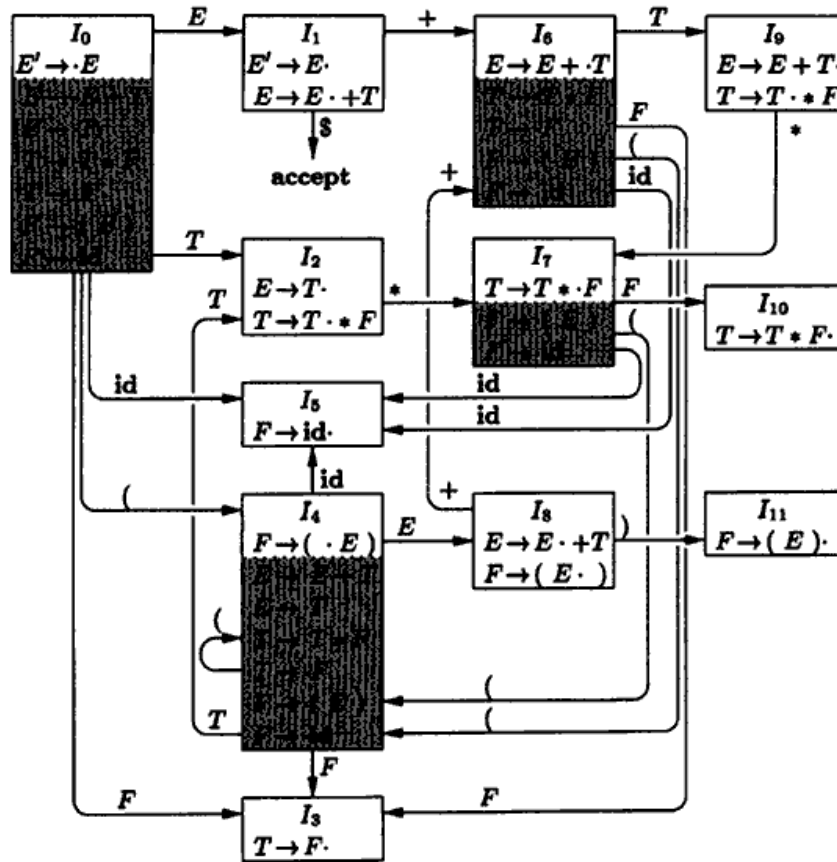
Q3. Solve the following.

A) Explain Operator Precedence parsing with an example.

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B) Consider the following LR canonical automaton of augmented grammar and Construct the LR parsing table.

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$$\begin{aligned} E' &\rightarrow E \\ E &\rightarrow E + T \mid T \\ T &\rightarrow T * F \mid F \\ E &\rightarrow (E) \mid id \end{aligned}$$


Q4. Solve Any Two of the following.

A) Explain Syntax Directed Definition with Inherited and Synthesized attributes.

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B) Construct the SDD and draw the annotated parse tree for the following string

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$(9+8*(7+6)+5)*4$ by computing its value when grammar is

$L \rightarrow E \quad E \rightarrow E+T \mid T \quad T \rightarrow T*F \mid F \quad F \rightarrow (E) \mid \text{digit}$

C) Explain dependency graph with an example.

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Q5. Solve Any Two of the following.

A) Explain Short-circuit code and Flow-of-Control statements in control flow.

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B) Construct the Directed Acyclic Graph, Thress Address Code with quadruples, triples for the following expression

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$((x+y)-((x+y)*(x-y)))+((x+y)*(x-y))$

- C) Explain type checking and type conversion with an example. 6
- Q6. Solve Any Two of the following.
- A) Explain optimization of basic blocks in brief. 6
- B) Explain the role of code generator with block diagram. 6
- C) Explain basic blocks and flow graphs with an example. 6

***** End *****